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Documenting and communicating the contributions of embedded statisticians: Show me the value!

Terrie Vasilopoulos[1,2] | Amy Crisp[3] | Gerard Garvan[3] | Keith Howell[1] | Gregory Janelle[1,4] | Cynthia Garvan[1]

[1]Department of Anesthesiology, College of Medicine, University of Florida, Gainesville, Florida, USA

[2]Department of Orthopaedic Surgery and Sports Medicine, College of Medicine, University of Florida, Gainesville, Florida, USA

[3]Center for Data Solutions, College of Medicine, University of Florida–Jacksonville, Jacksonville, Florida, USA

[4]Department of Surgery, College of Medicine, University of Florida, Gainesville, Florida, USA

Correspondence
Terrie Vasilopoulos, Department of Anesthesiology, College of Medicine, University of Florida, Gainesville, FL, USA.
Email: tvasilopoulos@anest.ufl.edu

Academic research productivity relies upon the contribution of statisticians, who are typically clustered in statistics and biostatistics departments, isolated from clinical researchers. Most academic health centres have created consultation hubs or research incubators to make statisticians available for individual collaboration to support the clinical research enterprise. Additionally, some clinical departments within academic health centres have recognized the value in colocating statisticians within their clinical departments to improve availability for collaboration with physicians/researchers. Embedded statisticians encounter the same challenges of isolated statisticians regarding professional support and networking, mentorship and clear role expectations. While for all collaborative statisticians, it is important to effectively communicate value to both collaborators and supervisors, this may be especially problematic for embedded statisticians in clinical departments where their supervisors may not have backgrounds in research or statistics. Previous papers have reported valuable metrics for statisticians, particularly those associated with Biostatistics, Epidemiology and Research Design Cores. There is a knowledge gap regarding metrics tailored to meet the needs of the embedded statistician and clinical supervisors. This paper is a first step towards addressing this important need.

In this paper, we explore (1) the critical role of collaborative statisticians and the benefits and challenges of the embedded statistician model, (2) the need for additional metrics specific to embedded statisticians which measure value and (3) how to design a value report. We offer a framework for evaluation of the contributions of the embedded statistician with the following domains: (1) collaboration, (2) research output/productivity, (3) mentoring and (4) education. Metrics that are particularly specific to embedded statisticians and that are not routinely captured include time from project initiation to completion/outcome, time from initial statistical consultation to statistical outcome completion and summary of level of contribution for manuscripts and presentations in addition to author order. We conclude with thoughts on future directions for development of metrics and reporting systems for statisticians embedded in clinical departments.

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1 | INTRODUCTION

Collaborative statisticians are integrated into academic research in a number of different ways. Many statistics and biostatistics departments within universities have found success creating a "consulting hub" that offers broad services to those requesting them. Some faculty within those departments also grow to be subject matter experts in other domains through long-term collaborations with outside researchers. There are also other collaborative statisticians who operate entirely outside of mathematical or statistical departments but who are still within the broader academic structure. Within the group of collaborative statisticians, there exist isolated statisticians (Figure 1), those outside structured statistics-related departments or statistical consulting hubs. Isolated statisticians face many challenges in terms of professional support and networking, mentorship and clear role expectations (Mehta et al., 2022). It is worthwhile to note that "isolated" is a relative term. A small consulting unit with one master's level analyst and one Ph.D. level statistician could easily be argued to contain two isolated statisticians.

Embedded statisticians (Figure 1) are isolated statisticians who are employed by other domain-specific departments where there is a sufficient volume of research being performed to justify an in-house statistician or analyst. In academic health centres, 18% of statisticians reported primary appointments in clinical departments rather than biostatistics or statistics/data science departments (Hanlon et al., 2022). There are both advantages and disadvantages for statisticians to be embedded in a clinical department (Spratt et al., 2017). One key advantage is the close interactions and long-term collaborations that can develop between embedded statisticians and clinical domain experts (Spratt et al., 2017). From these collaborations, embedded statisticians can find tremendous professional fulfilment, with projects related to improving patient care and clinical outcomes (Spratt et al., 2017).

However, embedded statisticians encounter the same challenges of isolated statisticians regarding professional support and networking, mentorship and clear role expectations (Mehta et al., 2022). Moreover, embedded statisticians are burdened by the requirement to communicate their needs, expectations and values to supervisors who are not statisticians and sometimes are also not researchers, within the structure and policies of their home clinical department; departmental-specific policies and structure can have a significant influence upon embedded statisticians' success and promotion (Spratt et al., 2017). These burdens are experienced by faculty and staff at every degree level. The metrics that have been previously proposed for collaborative statisticians often assume that the same language is being spoken between the statistician and supervisor (Griffith et al., 2022; Hanlon et al., 2022; Rubio et al., 2011) and may not encapsulate the activities performed by an embedded statistician. Thus, there is a knowledge gap regarding metrics tailored to meet the needs of the embedded statistician and clinical supervisor. In this paper, we will review the contributions of collaborative statistics, and embedded statisticians in particular, to the entire research enterprise; discuss perception of value by statisticians; outline metrics needed for embedded statisticians to communicate their value; and finally, provide examples on how embedded statistics can create value reports to share with their supervisors. Specifically, we will explore (1) the critical role of collaborative statisticians and the benefits and challenges of the embedded statistician model, (2) the need for additional metrics specific to embedded statisticians which measure value and (3) how to design a value report. We will conclude with thoughts on future directions for development of metrics and reporting systems for statisticians embedded in clinical departments.

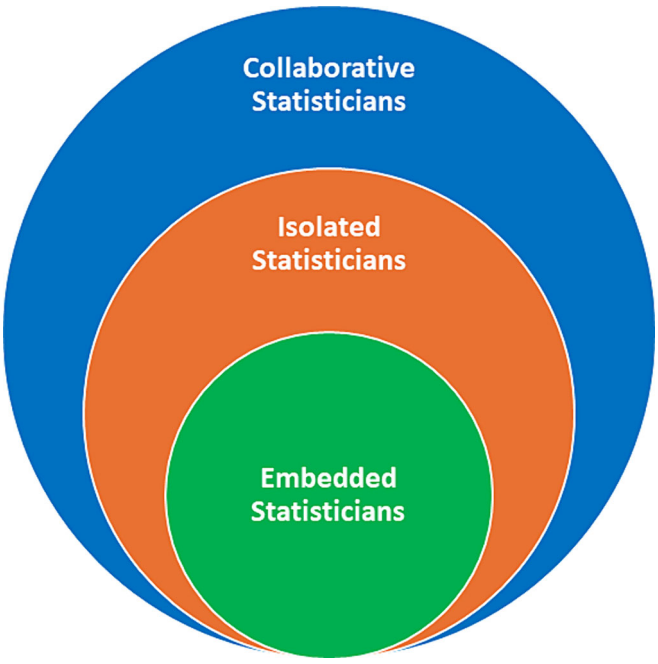


FIGURE 1 Relationships among collaborative, isolated and embedded statisticians.

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2 | CRITICAL ROLE OF COLLABORATIVE STATISTICIANS AND BENEFITS AND CHALLENGES OF THE EMBEDDED STATISTICIAN MODEL

For those trained in statistics and data science, there is an appreciation for the roles that statisticians play in every step in the entire investigative cycle from stating the research question, to trial design, data cleaning, performing the analyses and final interpretations of the results, all of which benefit from a trained analyst (Wild & Pfannkuch, 1999). These diverse roles necessitate that collaborative statisticians perform activities that cover a broad spectrum, with these activities varying across level of need and complexity of projects (Griffith et al., 2022). In clinical departments, the amount of research training and experience that clinicians have can vary from fairly modest to rather extensive and can depend on the specialty. An advantage for clinical departments with embedded statisticians is being able to have statistical input early and throughout projects. This early and in-depth involvement in projects can improve rigour, reproducibility and data integrity. Furthermore, embedded statisticians gain in-depth knowledge of their domain-specific field and can bridge gaps between clinical and statistical perspectives (Mehta et al., 2022). Greater content knowledge then allows embedded statisticians to not only construct more informed statistical models but also interpret results and better communicate findings to clinical audiences. Unfortunately, in other domains, there is the persistent idea that collaborative statisticians are simply data analysts and that the job begins after the data have been collected and ends when a *p*-value is delivered. When this is the case, how does one show that the large spans of time spent on tasks that seem ancillary are not only necessary but valuable?

3 | EXISTING MODELS FOR VALUE METRICS AND DOCUMENTATION

Previous publications have outlined guidelines and metrics to best document the work and value of collaborative statisticians (Griffith et al., 2022; Hanlon et al., 2022; Mazumdar et al., 2015; Rubio et al., 2011; Sharp et al., 2016; Spratt et al., 2017). A key component of these guidelines is the emphasis on the value of contributions by collaborative statisticians across multiple domains. While collaborative statisticians most recognizably contribute to the analysis and application of statistical methods, they also make substantial and impactful contributions to study design and implementation, manuscript/presentation/grant preparation, teaching and mentoring and service. Another critical component introduced in the Rubio et al. (2011) and Mazumdar et al. (2015) papers is the estimation of degree of involvement/effort related to activities. Rubio et al. (2011) advocated for the use of an ordinal metric (small, medium and large) that relates to the hours of effort required for a project (<10, 10–50 and 50+ h, respectively). In the Mazumdar et al. (2015) paper, they developed a framework to assess the scientific contributions of the collaborative biostatisticians, classifying contributions as minor, medium or major; their paper gives example of activities that would align with these different classifications. This type of effort appraisal will not only give better context to the contributions of embedded statisticians but would also give their clinical supervisors and administrators a metric to evaluate workload and effort distribution.

4 | METRICS FOR EMBEDDED STATISTICIANS

While many metrics for collaborative statisticians apply to embedded statisticians, statisticians embedded in clinical departments also have unique evaluation needs. Spratt et al. (2017) noted that the working environment in clinical departments differs from that of a statistics or biostatistics department in that patient care is, rightly so, the priority. Overall, the ability of embedded statisticians to contribute to scholarly activity is highly dependent on the schedules and priorities of their clinical collaborators, meaning that scholarly productivity for embedded statisticians can be more affected by opportunity than their skills as a collaborative statistician. The primacy of patient care can slow the research process and, in turn, the rate of publications for a collaborative statistician. Furthermore, the emphasis on writing and attaining grant funding for clinical faculty is lower, especially if they themselves are not on a tenure track. However, the focus on patient care also provides an opportunity for embedded statisticians to demonstrate their value by showing how their contributions to clinical projects have improved health outcomes and hospital processes.

Additionally, teaching opportunities and responsibilities for statisticians embedded in clinical departments can vary and be irregular. In clinical departments, there is more emphasis on one-on-one interactions compared with teaching in more formal courses (Spratt et al., 2017). Likewise, many clinical departments do not have affiliated graduate degree programmes, with mentoring activities focusing on medical students, residents, fellows and other faculty (Spratt et al., 2017). If more formal teaching and graduate committee work is desired by an embedded statistician, they will need to seek out appointments in other departments and colleges, as well as communicate and negotiate this effort in their home clinical department.

Furthermore, many metrics that are underemphasized and prioritized in other fields can be of great importance to statisticians embedded in clinical departments. In a survey of leaders of collaborative biostatistics units in academic health centres, Hanlon et al. (2022) found that metrics appraised as the most important in terms of career advancement were coinvestigator status on NIH or similar grants as well as first or second authored peer-reviewed publications, while middle authorship was ranked as one of the least important factors. However, collaborative statisticians, especially those embedded in clinical departments, often have increased middle authorship papers, with less emphasis on first authorship.

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Additionally, co-authorship as a collaborative biostatistician typically still represents a substantial contribution to publications (Sharp et al., 2016). This is why documenting degree of effort, as mentioned above, is extremely important. Moreover, for projects with planned extensive statistical work or modelling, collaborative statisticians should negotiate and advocate for a higher authorship listing.

Additionally, collaborative biostatisticians—and embedded statisticians in particular—tend not to have their own research programme in the traditional sense but instead are more likely to have a statistical "theme," that is, a set of specialty skills and tools that they can apply to their domain-specific fields. Due to their expertise and vital skills, embedded statisticians often are involved on many projects at once, linked not by the research topic but more by the specific statistical needs contributed by the statistician. Furthermore, being involved in multiple projects, and with multiple collaborators, should be viewed as a positive for a collaborative statistician, supporting their well-rounded abilities as well as their reputation as a collaborator (Griffith et al., 2022). Table 1 provides a framework to highlight which of the metrics previously discussed are most useful to embedded statisticians as well as suggest additional metrics to be considered, with metrics included with appreciation for both clinical department relevance and ease of data capture. There are four domains in the metric framework: (1) collaboration, (2) research output/productivity, (3) mentoring and (4) education. Metrics that are particularly specific to embedded statisticians and that are not routinely captured include time reporting for both project initiation to completion/outcome and initial statistical consultation to statistical outcome completion. These variables will account for the often longer timelines needed to publish with clinical collaborators. Additionally, the authors strongly emphasize the need to report not only author order of manuscripts and presentation but also level/context of contribution. Level of contribution can vary widely and may include supervising trainees' statistical analyses, writing the majority of the methods and results sections, leading the study design plan from onset and/or being an integral coinvestigator on related grant funding for project. We encourage embedded statisticians to develop a list with descriptors of levels of contribution that occur most frequently in their collaborations so that integrating these metrics into reports can be streamlined.

TABLE 1 Proposed metrics for embedded statisticians.

Domain	Metrics
Collaboration	Number of projects • Effort estimation: Small, medium or large Number of unique collaborators • Classified by division/lab/area and so forth • Time from project initiation to completion/outcome (e.g., manuscript) • Time from initial collaboration to collaborative statistical outcome completion (e.g., analysis completion and results write-up)
Research output/ productivity	Number of manuscripts • In preparation • Submitted • Published • Journals ◦ Note authorship order (first, last, second, or middle) ◦ Summarize extent of contribution Number of presentations • Type (poster, oral, etc.) • Conference ◦ Note authorship order (first, last, second, or middle) Number of grants • Submitted • Under review • Newly awarded • Ongoing ◦ Note grant PI and own role (e.g., Co-I), funding agency and total amount
Mentoring	Number of mentees • Training level • Type of mentoring activities • Time devoted to mentoring (e.g., number of meetings per year) • Involvement in career development awards • Mentee outcomes (e.g., presentation, publications and promotions)
Education	Number of educational activities • Type (e.g., lecture, workshop, course and small group) • Affiliation (department, college, etc.) • Level of learners • Number of learners • Time/effort for activities (e.g., length of activities, assignments and grading) • Learner evaluations • Peer evaluations

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5 | DESIGNING A VALUE REPORT

While defining metrics for embedded statisticians is important, it is just as important for embedded statisticians to be able to develop reports to best communicate theory work and value with their nonstatistician supervisors. Unfortunately, there is not a standardized way to generate reports to communicate one's value. These authors emphasize that embedded statisticians need to take responsibility for these reports and control their own narrative. Reports or work summaries generated by departmental administration can lack description of both the breadth and depth of activities performed by embedded statisticians, often focusing on small snapshots of their work (e.g., number of publications and external funding). There are several important factors for embedded statisticians to consider when tracking metrics to develop their report:

1. Content: What metrics? (see above)
2. Process: How do we document research workflow and involvement at each stage?
3. Software: What software should we use to track metrics and easily generate reports?
4. System: How do we organize and classify the metrics to best reflect our activities?

It is both necessary and beneficial for embedded statisticians to meet and discuss what information is useful to their supervisors, because the approaches to the above will vary depending on the department as well as on the individual statistician. Figure 2 presents a sample value report for an embedded statistician in a clinical department that was created in consultation with the clinical departmental chair. Within the authors' clinical department, we are piloting the use of these reports for quarterly productivity updates, which has spurred increased communication between embedded statisticians and not only the department chair but also with other departmental leaders (i.e., divisions chiefs and senior administrators). These reports also augment existing quarterly research reports from the finance office on grant funding for faculty by highlighting information not captured by institutional financial reporting, particularly grants in which a faculty member is a coinvestigator or other personnel and not the principal investigator. Utilizing visualizations helps create easy to understand value reports that focus on maximizing communication and understanding.

However, while it is important for embedded statisticians to generate these reports, the system needs to be streamlined with minimal time and effort to be maintained by the statistician. This will ensure that the reporting system is sustainable and does not contribute to burnout. Again, this is why close communication with supervisors is necessary to align expectations. One way to streamline report generation is to align and automate components when possible. This can include having set time periods for reporting (e.g., quarterly), utilizing existing institutional infrastructure to obtain needed information (e.g., grants offices) and taking advantage of existing features of websites to gain information (e.g., PubMed alerts). Furthermore, many available task and project management software programs have customizable automation to easily classify activities and track time spent on projects. A more advanced step would be to develop specialized, interactive tools to track and summarize metrics. An institutional case study (Schmidt et al., 2017) demonstrated that a web-based dashboard application for research project tracking in an epidemiology/biostatistics department, using REDCap and R programming, reduced study intake time, increased efficiency and improved workload allocation. Similar integrations should be used to develop dashboards to track metrics and research workflows to better—and more efficiently—summarize activities and generate value reports for embedded statisticians.

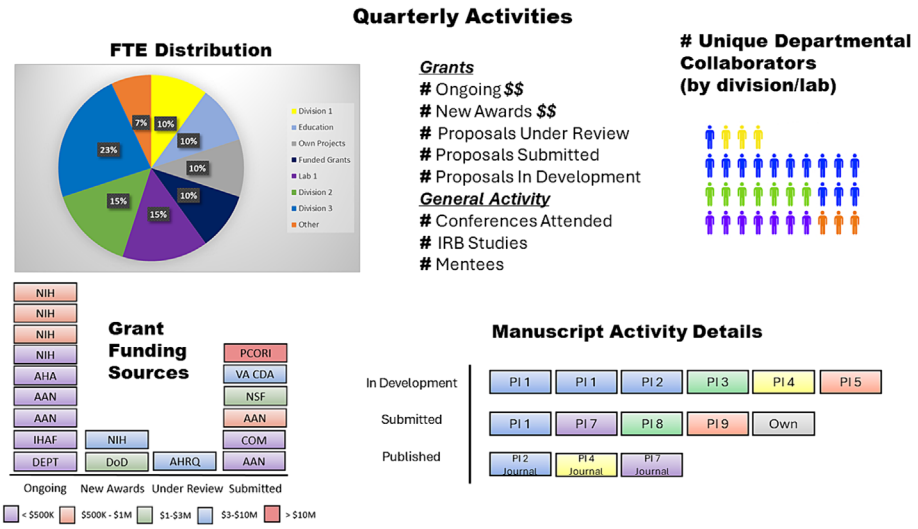


FIGURE 2 Sample value report. Report contains example quarterly summaries for FTE distribution by division/lab/other activities; number of unique collaborators; manuscripts (in preparation, submitted and published); grant funding (ongoing, new, under review and submitted) with classification by funding agency; and other activities including IRB studies as Co-I, conferences attended and mentees.

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6 | FUTURE DIRECTIONS

The next step for development of metrics and reporting systems for statisticians embedded in clinical departments is additional study of value perceptions. Thus far, to our knowledge, the population of embedded statisticians, their clinical collaborators and their clinical supervisors has not been studied. Recent surveys of collaborative statisticians have evaluated perceptions on value and recognition by colleagues, departments and institutions, with these studies comparing collaborative statisticians with and without statistician/methodologist supervisors. Sharp et al. (2016) conducted a survey of statistical consultants in university settings to assess perceived value. Twenty-seven respondents were within established consulting centres while 12 were not. Statisticians not in consulting centres did not have significant differences in rates of agreement, compared with those in consulting centres, in terms of being valued by the institution (50% vs. 67%), by faculty outside department (83% vs. 96%) and by faculty in their department (58% vs. 74%), though these observed rates were slightly lower. There were also similar agreement rates reported between statisticians not in a consulting centre versus those within viewing that their work contributed to promotion (58% vs. 56%) and publication requirements (67% vs 69%). In another survey focussed on collaborative biostatisticians in academic health centres, the majority (87%) of respondents thought that their work was valued by their collaborators (Devick et al., 2022). Encouragingly, this rate was similar for statisticians with (87%) and without (82%) supervisors who were methodologists (Devick et al., 2022).

However, these studies also noted differences for statisticians not in a consulting centre or collaborative subunit or those without a methodology supervisor. Sharp et al. (2016) found that statisticians not in a consulting centre were more likely (83%) to report a higher consulting workload relative to other responsibilities compared with statisticians in a consulting centre (63%). Additionally, statisticians not in a consulting centre were less likely to agree that their work contributed to pay raises (42%) compared with those who were in consulting centres (93%). Devick et al. also reported that statisticians without a methodologist supervisor, compared with those with one, were less likely to report the ability to collaborate with other statisticians or other researchers with quantitative backgrounds (33% vs. 62%) and the opportunity to collaborate specifically on methodological research (15% vs. 39%).

The work of Sharp et al. and Devick et al. needs to be replicated and extended to the population of embedded statisticians, their clinical collaborators and their clinical supervisors. Such work would include a survey of extant metrics and reporting systems used in clinical departments and the value perceptions of clinical faculty and leadership. Importantly, research is needed on the economics of the embedded statistician model to answer questions such as cost effectiveness and contribution of embedded statisticians to physician promotion, retention and well-being.

7 | CONCLUSION

We have provided here the context for understanding the unique challenges of embedded statisticians within the larger academic research paradigm and, in particular, the need for these individuals to advocate for themselves in a way that translates the full extent of their value to their supervisors. We have also given examples of metrics that we have found useful in our own efforts to communicate with such individuals. We hope this helps to guide a continued discussion on methods and practices for supporting isolated and embedded statisticians in this rapidly growing field.

AUTHOR CONTRIBUTIONS

Terrie Vasilopoulos: Conceptualization; writing—original draft; writing—review and editing. **Amy Crisp:** Conceptualization; writing—original draft; writing—review and editing. **Gerard Garvan:** Conceptualization; writing—original draft; writing—review and editing. **Keith Howell:** Conceptualization; writing—original draft; writing—review and editing. **Gregory Janelle:** writing—original draft; writing—review and editing. **Cynthia Garvan:** Conceptualization; writing—original draft; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

None.

DATA AVAILABILITY STATEMENT

This is a narrative review and no data were used.

ORCID

Keith Howell https://orcid.org/0009-0009-7045-6621

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AUTHOR BIOGRAPHIES

Dr. Terrie Vasilopoulos is a collaborative biostatistician and an associate professor in the Departments of Anesthesiology and Orthopaedic Surgery & Sports Medicine at the University of Florida College of Medicine and serves as chief of the Department of Anesthesiology's Division of Education Research and Scholarship. She holds a PhD in Biobehavioral Health from the Pennsylvania State University and Graduate Statistician Certification from the American Statistical Association.

Dr. Amy Crisp is a collaborative biostatistician and the associate director of the Center for Data Solutions at the University of Florida College of Medicine—Jacksonville. She holds a PhD and an MS in Biostatistics from University of Florida and an MS and a BS in Physics from Florida State University.hD in Biobehavioral Health from the Pennsylvania State University and Graduate Statistician Certification from the American Statistical Association.

Gerard Garvan MS, is a biostatistician in the Center for Data Solutions at the University of Florida College of Medicine–Jacksonville. He holds an MS in Biostatistics from the University of Florida and a BS in Mathematics from the University of Florida.

Dr. Keith Howell is an associate professor in the Department of Anesthesiology at the University of Florida College of Medicine and serves as chief of the Department of Anesthesiology's Division of Clinical Informatics. He earned his MD at the Medical College of Georgia in Augusta, completed a surgical internship at Carraway Methodist Medical Center in Birmingham, AL, and completed anesthesiology residency at the Virginia Commonwealth University School of Medicine.

Dr. Gregory Janelle is a professor in the Departments of Anesthesiology and Surgery at the University of Florida College of Medicine and serves as interim chair of the Department of Anesthesiology. He is a recognized fellow of the American Society of Echocardiography and the American Society of Anesthesiologists. He earned his MD at Boston College and completed anesthesiology residency and a cardiothoracic anesthesiology fellowship at the University of Florida College of Medicine.

Dr. Cynthia Garvan is a collaborative biostatistician and professor in the Department of Anesthesiology at the University of Florida College of Medicine and serves as co-director of the College of Medicine's Translational Research Academy for Clinical Translational Science. She holds a PhD in Biostatistics from the University of Florida and an MS in Mathematics from the Pennsylvania State University.

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